Part 1: Analytics Design

System Performance

	Question	Data source
1	How does the number of transcript segments per conversation vary, i.e. are most conversations short or long?	TranscriptSegment.conversation count per Conversation
2	Which conversations have the most text input (word count)?	TranscriptSegment.text word count
3	How consistent is the AI's note-generation output across conversations of similar length?	ImprovementNote count per conversation grouped by segment-count bucket

User Behavior

	Question	Data source
1	How frequently are conversations being recorded or what does daily/weekly usage look like??	Conversation.recorded_at
2	What types of improvement feedback does the AI generate most from filler words, pace, tone, confidence	ImprovementNote.note_type frequency
3	How many conversations have had improvement notes generated vs. none i.e	Conversation count split by whether any ImprovementNote exists for it

Cost

	Question

1	How does estimated processing cost vary across users?
2	What is the average cost per session, and how has it trended over time?
3	How does estimated cost (word count) vary across conversation duration buckets like short, medium, long?

Step 2: Dashboard Planning

Widgets:

Question	Widget	Data Required
SP-1: Segment count distribution	Histogram	TranscriptSegment count per Conversation
SP-2: Top conversations by word count	Horizontal bar chart	Word count of TranscriptSegment.text per Conversation
SP-3: Note consistency by conversation length	Grouped bar chart	ImprovementNote count grouped by segment-count bucket (short/medium/long)
UB-1: Daily/weekly usage	Line chart	Conversation.recorded_at grouped by day
UB-2: Most common note types	Horizontal bar chart	ImprovementNote.note_type frequency
UB-3: AI adoption rate	Donut chart	Conversation count split by whether any ImprovementNote exists
C-1: Cost per user	Stacked bar chart	Total word count per Conversation grouped by user
C-2: Average cost over time	Line chart	Mean word count per conversation grouped by week
C-3: Cost by duration bucket	Grouped bar chart	Average word count grouped by duration_seconds bucket (<60s, 60–180s, 180s+)

Step 3: Data Science Thinking

- **Distributions** (SP-1, SP-3): histograms reveal shape (skewed vs. uniform) that averages alone hide
- **Correlation** (SP-2, C-3): tests whether longer input produces proportionally more output/cost
- **Trends over time** (UB-1, C-2): 7-day rolling average smooths noise to surface genuine usage growth or decline
- **Group comparison** (SP-3, C-3): bucketing conversations by length shows whether AI behavior is consistent across input sizes
- **Proportions vs. counts** (UB-3): donut chart adds relative view alongside raw numbers for a complete picture
- **Cost proxy reasoning** (C-1, C-2, C-3): word count is used as a proxy for compute cost since token count scales linearly with word count — relative comparisons between users and sessions remain valid even without billing data

Part 2:
Screenshots:

EchoLabs Dashboard Analytics Progress Insights Transcribe Conversations Rep

Analytics Dashboard

System performance, user behavior, and cost — across every recorded conversation in EchoLabs. Cost is approximated from word count using a $0.0000026/word proxy rate.

CONVERSATIONS	USERS	TOTAL SEGMENTS	AVG SEGMENTS / CONV
150	6	732	4.9
IMPROVEMENT NOTES	**TOTAL WORDS**	**TOTAL EST. COST**	**AVG COST / CONV**
179	9552	$0.0248	$0.0002

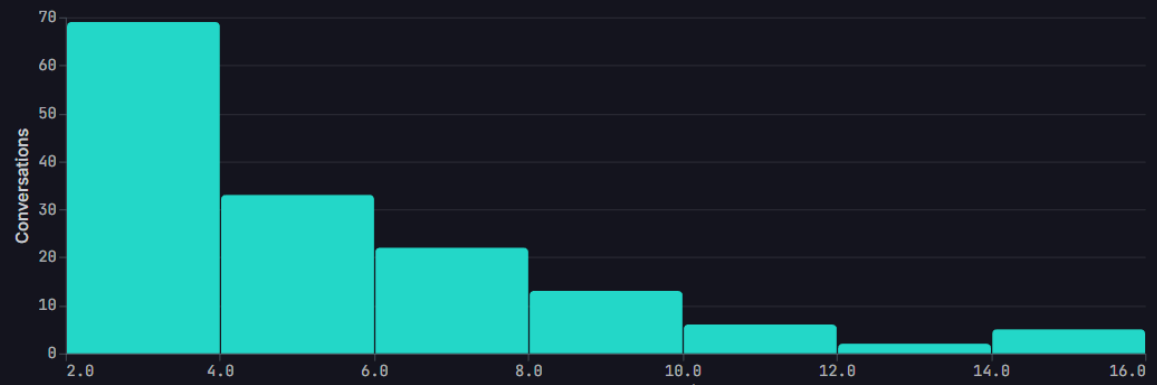
EchoLabs Dashboard Analytics Progress Insights Transcribe Conversations Re

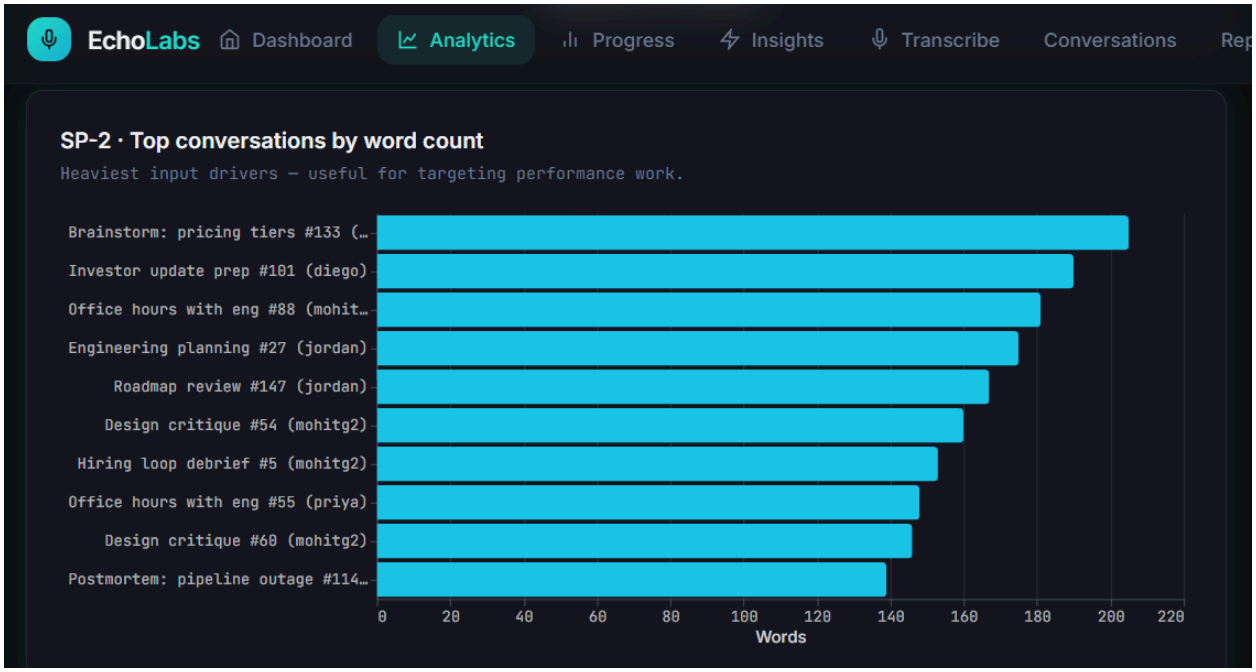
System Performance

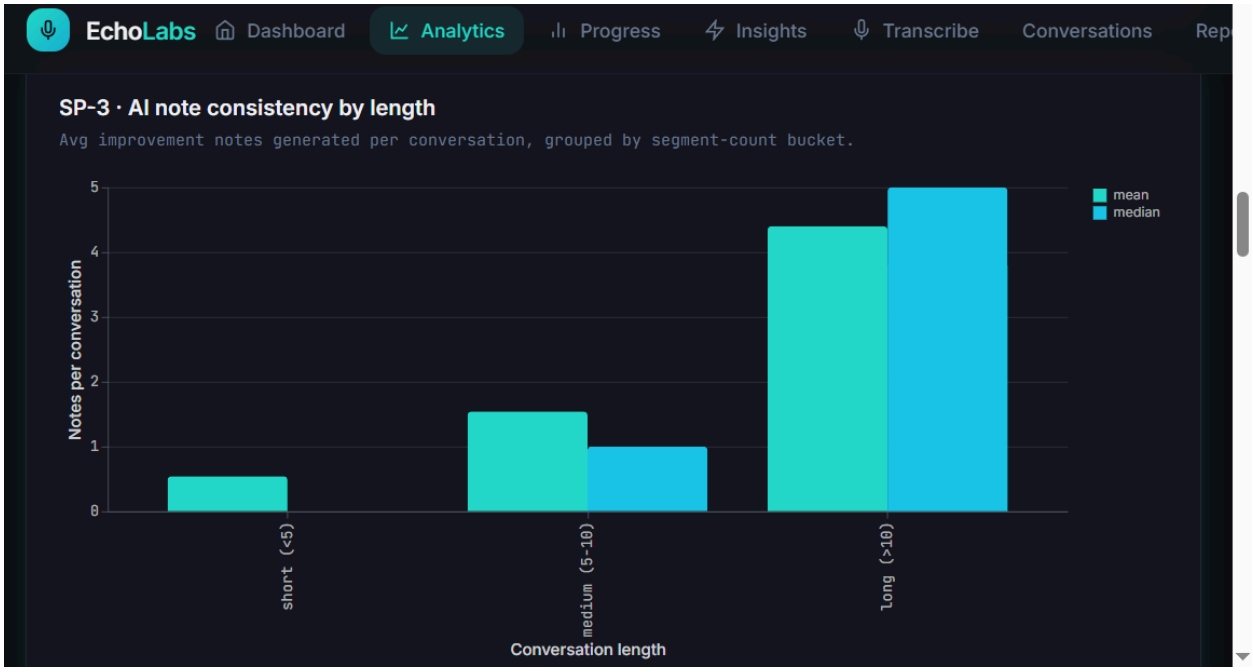
How long are conversations? Which inputs dominate? Is AI output consistent across input sizes?

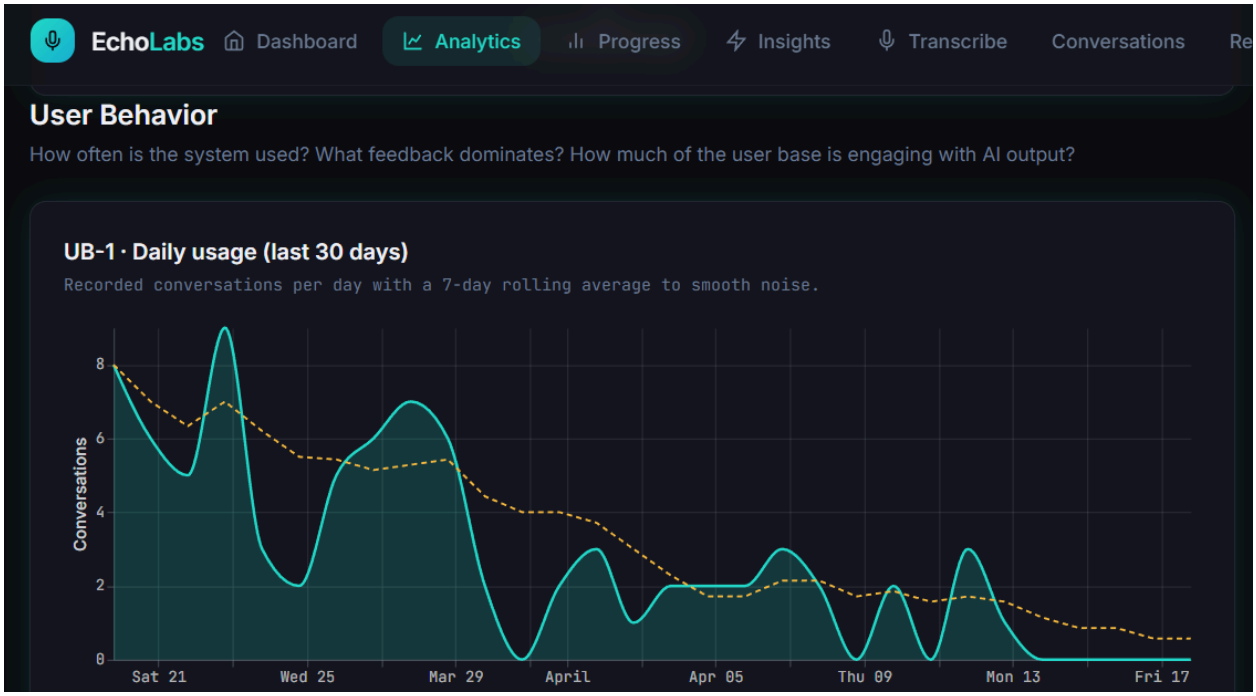
SP-1 · Segment-count distribution

Histogram of segments per conversation — exposes skew between short and long sessions.

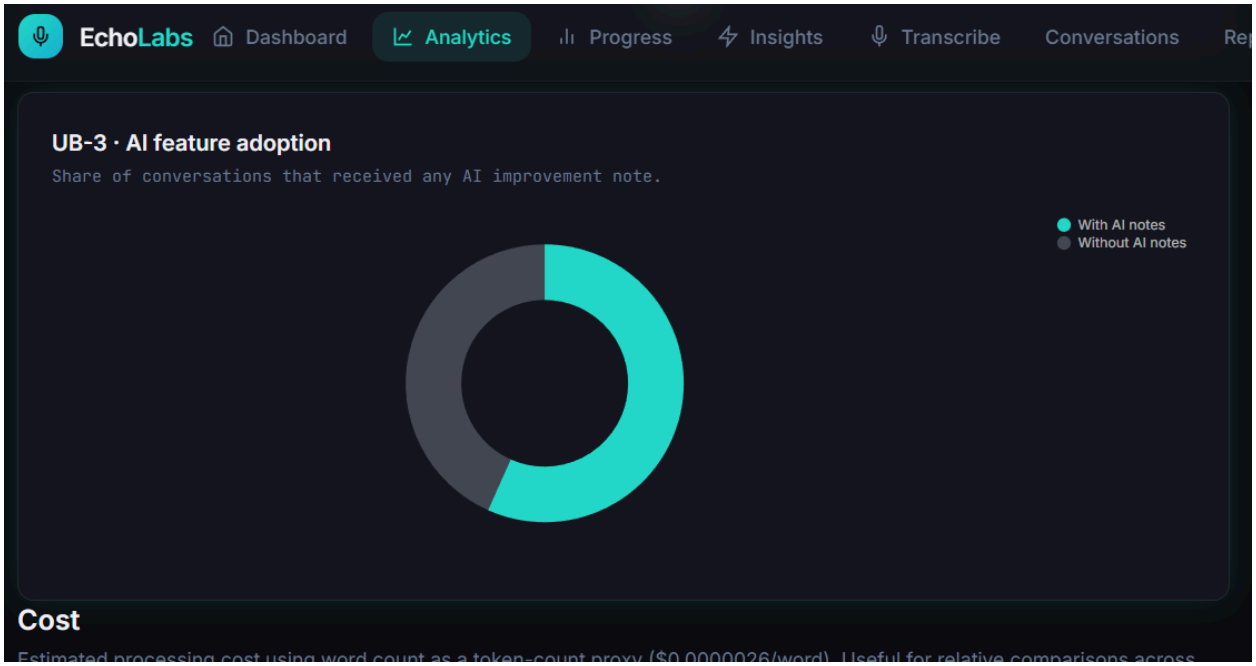


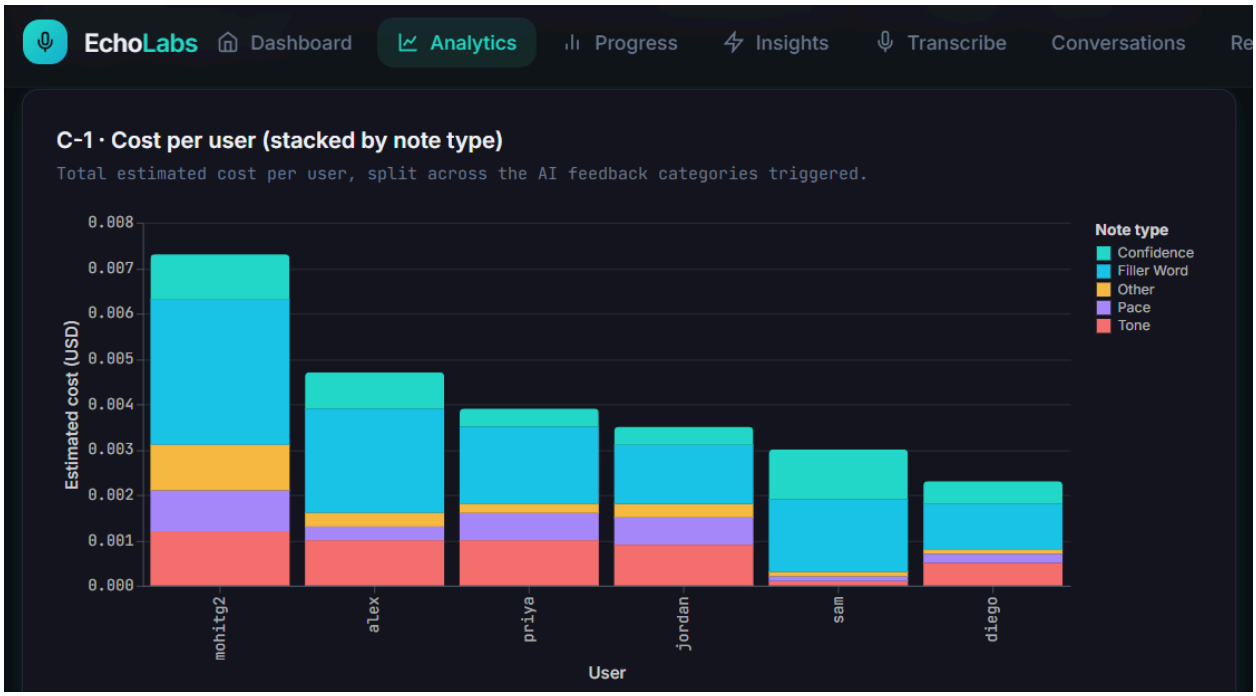

EchoLabs
Dashboard
Analytics
Progress
Insights
Transcribe
Conversations
SP-2 · Top conversations by word count
Heaviest input drivers — useful for targeting performance work.
Brainstorm: pricing tiers #133 (…
Investor update prep #101 (diego)
Office hours with eng #88 (mohit…
Engineering planning #27 (jordan)
Roadmap review #147 (jordan)
Design critique #54 (mohitg2)
Hiring loop debrief #5 (mohitg2)
Office hours with eng #55 (priya)
Design critique #60 (mohitg2)
Postmortem: pipeline outage #114…
0
20
40
60
80
100
120
140
160
180
200
220
Words

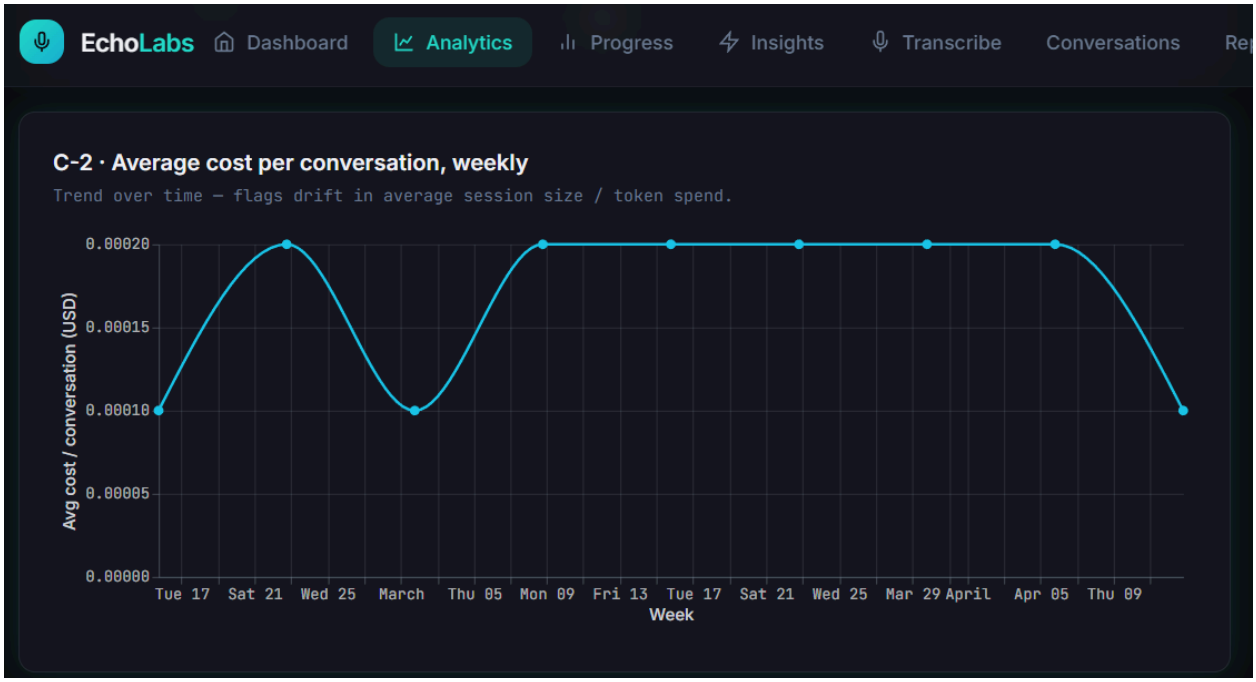

EchoLabs
Dashboard
Analytics
Progress
Insights
Transcribe
Conversations
SP-3 · AI note consistency by length
Avg improvement notes generated per conversation, grouped by segment-count bucket.
mean
median
Notes per conversation
0
1
2
3
4
5
short (<5)
medium (5-10)
long (>10)
Conversation length

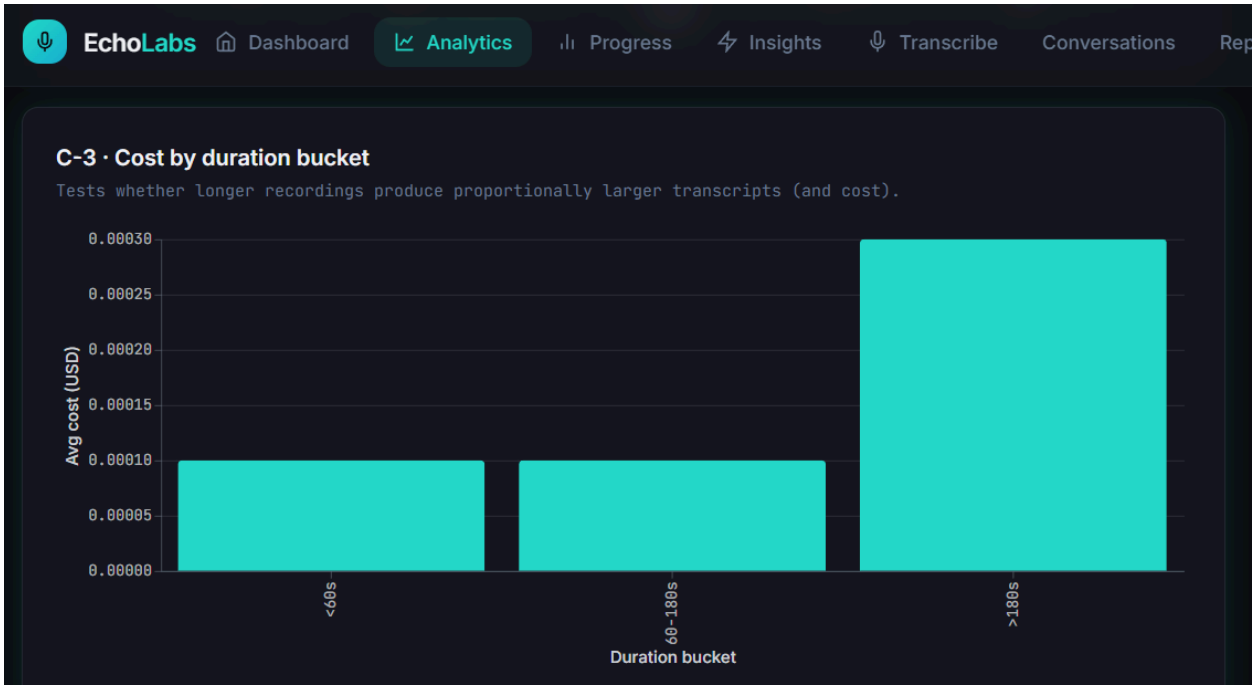

EchoLabs
Dashboard
Analytics
Progress
Insights
Transcribe
Conversations
Re
User Behavior
How often is the system used? What feedback dominates? How much of the user base is engaging with AI output?
UB-1 · Daily usage (last 30 days)
Recorded conversations per day with a 7-day rolling average to smooth noise.
Conversations
0
2
4
6
8
Sat 21
Wed 25
Mar 29
April
Apr 05
Thu 09
Mon 13
Fri 17

EchoLabs
Dashboard
Analytics
Progress
Insights
Transcribe
Conversations
Rep
UB-2 · Most common note types
Frequency of filler / pace / tone / confidence feedback the AI generates.
Filler Word
Tone
Confidence
Pace
Other
0
5
10
15
20
25
30
35
40
45
50
55
60
65
70
75
80
Notes


EchoLabs
Dashboard
Analytics
Progress
Insights
Transcribe
Conversations
UB-3 · AI feature adoption
Share of conversations that received any AI improvement note.
With AI notes
Without AI notes
Cost


EchoLabs
Dashboard
Analytics
Progress
Insights
Transcribe
Conversations
C-1 · Cost per user (stacked by note type)
Total estimated cost per user, split across the AI feedback categories triggered.
Estimated cost (USD)
0.008
0.007
0.006
0.005
0.004
0.003
0.002
0.001
0.000
mohitg2
alex
priya
jordan
sam
diego
User
Note type
Confidence
Filler Word
Other
Pace
Tone


EchoLabs
Dashboard
Analytics
Progress
Insights
Transcribe
Conversations
C-2 · Average cost per conversation, weekly
Trend over time — flags drift in average session size / token spend.
Avg cost / conversation (USD)
0.00020
0.00015
0.00010
0.00005
0.00000
Tue 17
Sat 21
Wed 25
March
Thu 05
Mon 09
Fri 13
Tue 17
Sat 21
Wed 25
Mar 29
April
Apr 05
Thu 09
Week


EchoLabs
Dashboard
Analytics
Progress
Insights
Transcribe
Conversations
C-3 · Cost by duration bucket
Tests whether longer recordings produce proportionally larger transcripts (and cost).
Avg cost (USD)
0.00030
0.00025
0.00020
0.00015
0.00010
0.00005
0.00000
<60s
60-180s
>180s
Duration bucket